

SuperBox: A 4-in-1 Power Electronics Solution for Dual-Motor Electric Vehicles

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Abstract—The integration of power electronics systems in electric vehicles (EVs) is increasingly attractive for achieving compactness and cost-effectiveness. This paper introduces the SuperBox, a highly integrated 4-in-1 power electronics solution specifically designed for EV applications. The scope of SuperBox encompasses the functionalities of on-board charging, DC boost charging, motor traction, and auxiliary power modules (APMs), all within a single housing. While leveraging established topologies to facilitate rapid industry adoption, SuperBox is distinguished by its novel approach to functional integration. This integration is achieved through three key technical strategies, namely utilizing motor windings for charging, employing a three-port transformer to magnetically couple the high-voltage and low-voltage batteries while maintaining galvanic isolation, and repurposing the traction inverter as a totem-pole power factor correction (PFC) circuit in charging mode. Implementation of such features eliminates the need for the PFC stage of the on-board charger as well as the primary-side circuitry of the APM converter. Subsystem simulations demonstrate the efficacy of SuperBox to achieve significant cost savings while delivering performance on par with conventional systems. Preliminary benchmarking suggests a 15% saving in the power stage cost compared to conventional non-integrated solutions.

Index Terms—electric vehicle, integrated on-board chargers, power converter, triple active bridge, vehicular technology

I. INTRODUCTION

At the heart of every battery electric vehicle (BEV) is a complex network of power electronics, as depicted in Fig. 1. The power converters and their control systems convert and manage electrical energy to drive the vehicle, charge its battery, and optimize overall system performance [1], [2].

The current market demands highly integrated power electronics systems in the next generation of EVs. Reducing the size, cost, and weight of power electronics converters, such as the on-board charger (OBC), traction inverter, and auxiliary power module (APM), is a key requirement [3]. To meet this objective, integrated OBC (iOBC) has become popular, whereby other functionalities or subsystems that are conventionally not part of the OBC are integrated within the OBC package, as the name suggests [4], [5]. A basic level of integration refers to integrating multiple systems mechanically into a single housing, thus reducing the total housing material as well as the cold plate used to cool the electronics. For instance, the solution in [6] combines the APM—a 400-V/12-V converter—with the OBC in a single package along with incorporating vehicle-to-grid (V2G) charging functionality.

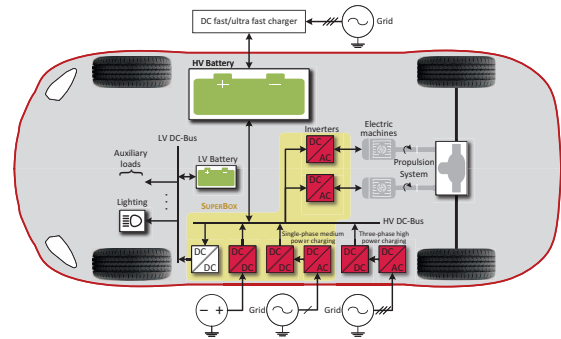


Fig. 1. Battery electric vehicle (BEV) powertrain showing SuperBox scope.

A higher degree of integration in iOBCs is commonly referred to as functional integration, wherein components within the high-voltage (HV) powertrain are repurposed to perform multiple functions [5]. Such iOBC topologies leverage the mutually exclusive nature of traction and charging operations (except in the case of wireless charging). The iOBC topologies can be classified into three categories: (1) utilizing the traction inverter as a charger, (2) employing the propulsion motor as an inductive element, and (3) magnetic integration using multi-winding high-frequency transformers.

In [7], DC fast charging is achieved by configuring a dual inverter drive system as a differential input to the HV DC input, where the two inverters are driving a single open-end winding machine (OEW). However, this necessitates the existence of a second inverter to facilitate the differential input. More importantly, the traditional OBC system must also exist to facilitate AC charging. AC and DC charging through an OEW was also proposed in [8]. The benefits of employing the traction inverter as a charger are most pronounced when the propulsion motor is repurposed. Specifically, using the propulsion motor as an inductive element eliminates the need for conventional input chokes in the power factor correction (PFC) stage of the OBC. In this case, the AC input is fed into the motor windings that are connected to the traction inverter. For instance, [9] proposes a DC fast charger where the DC input is fed into the motor windings connected to the traction inverter. The fast charger exploits the motor windings as a boost inductor with the addition of an extra half-bridge switching leg. However, this topology does not support AC

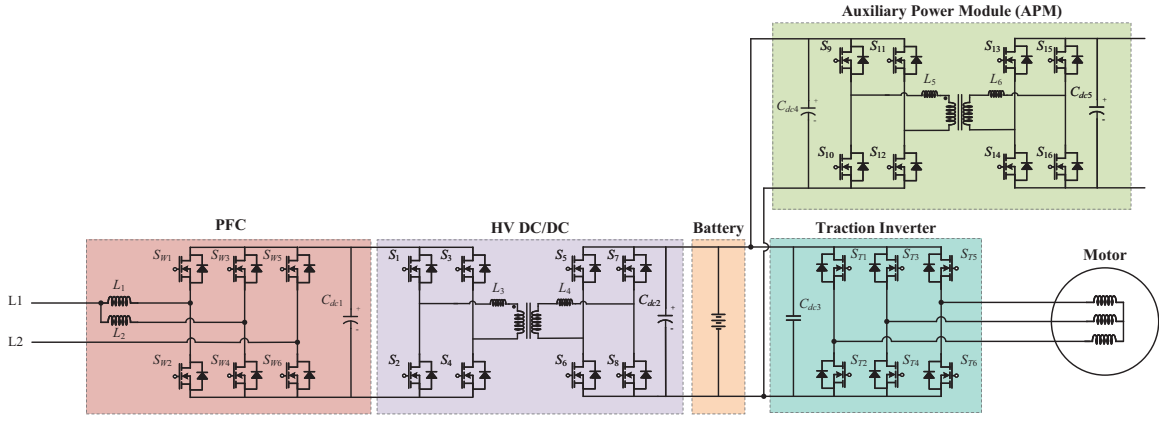


Fig. 2. Simplified diagram of the conventional power electronic system in BEVs.

charging. A single-phase AC charging iOBC is presented in [4]. However, the topologies in [9] and [4] deliver non-isolated solutions, which limits their application in scenarios where isolation is mandated by standards.

To overcome the isolation challenge, an iOBC topology with a six-phase motor is introduced in [10]. With the help of a triple-pole switch, the motor is repurposed as an isolation transformer. In this configuration, the terminals of one of the two three-phase inverters are disconnected and reconfigured as the AC input. The charging power is magnetically transferred to the second three-phase inverter, which is connected to the battery. Similar multiphase motor-based iOBCs approaches, with extension to nine-phase machines for three-phase AC charging, are discussed in [5] and [11]. However, the limited adoption of multiphase motors in the automotive industry nowadays presents a significant barrier to these implementations. Lastly, magnetic integration using multi-winding transformers enables the coupling of multiple power converters while maintaining galvanic isolation [12], [13]. For example, [14] demonstrates magnetic integration of the OBC, APM, and the on-board receiver pad of a wireless power transfer system. While the topology offers a high degree of integration, the traction drive is not considered.

This paper introduces SuperBox, a highly integrated 4-in-1 topology that combines the traction drive, OBC, DC boost charging, and APM. The high level of functional integration is achieved through three key strategies: (1) utilizing motor windings as components for charging, (2) employing a three-port transformer that magnetically couples HV and LV batteries, and (3) repurposing the traction inverter to function as a totem-pole power factor correction (PFC) circuit in charging mode. SuperBox also supports dual motor applications to provide high charging and traction power. This approach enables the seamless integration of a redundant APM for autonomous driving and functional safety purposes.

The remainder of this paper is organized as follows. Section II introduces the working principles of the SuperBox. Section III investigates some of the challenges posed by the high integration and discusses potential solutions. Sim-

ulation results are presented in Section IV. Benchmarking of the proposed SuperBox system against the conventional non-integrated solution is undertaken in Section V. Finally, Section VI concludes the paper.

II. POWERTRAIN ARCHITECTURE AND PROPOSED INTEGRATED OBC

A. Conventional System

Fig. 2 depicts the conventional power electronics system in BEVs, omitting safety and protection relays for simplicity. The AC input at terminals L1 and L2 is converted to DC by the PFC circuit and input chokes $L_{1,2}$, which also regulates the power factor (PF) of the grid. The HV DC/DC forms the second stage of the OBC, and outputs a voltage suitable for the HV battery while providing galvanic isolation, with semiconductor switches enabling bidirectional power flow. The APM supplies low-voltage (LV) power to auxiliary loads, while the traction inverter powers the AC motor from the HV battery during driving mode. DC boost charging can be enabled by bypassing the PFC stage with relays, provided that the HV DC/DC supports 1:2 voltage boosting.

The semiconductor carbon footprint in Fig. 2 is quite large with many switches sharing similar characteristics. The proposed SuperBox aims to reduce this footprint significantly.

B. SuperBox: Proposed iOBC Topology

Fig. 3 presents the diagram of the SuperBox. It consists of two traction inverters, two triple-active bridge (TAB) converters with a three-port transformer (3PT), and mode selector relays: S_M , S_P , and S_{DC} . The SuperBox is designed for an 800-V BEV powertrain, with at least two electric drive modules (EDMs). Thus, the subscripts *A* and *B* in Fig. 3 denote banks A and B for the two EDMs, respectively. The two EDMs could be the front and rear motors of a dual-motor BEV, such as the 2022 Hyundai IONIQ 5 [15]. Alternatively, they could also be the two rear motors of a quad-motor BEV, like the 2022 Rivian R1T [16].

By utilizing the motor windings, as input filters during AC and DC charging modes, and repurposing the traction

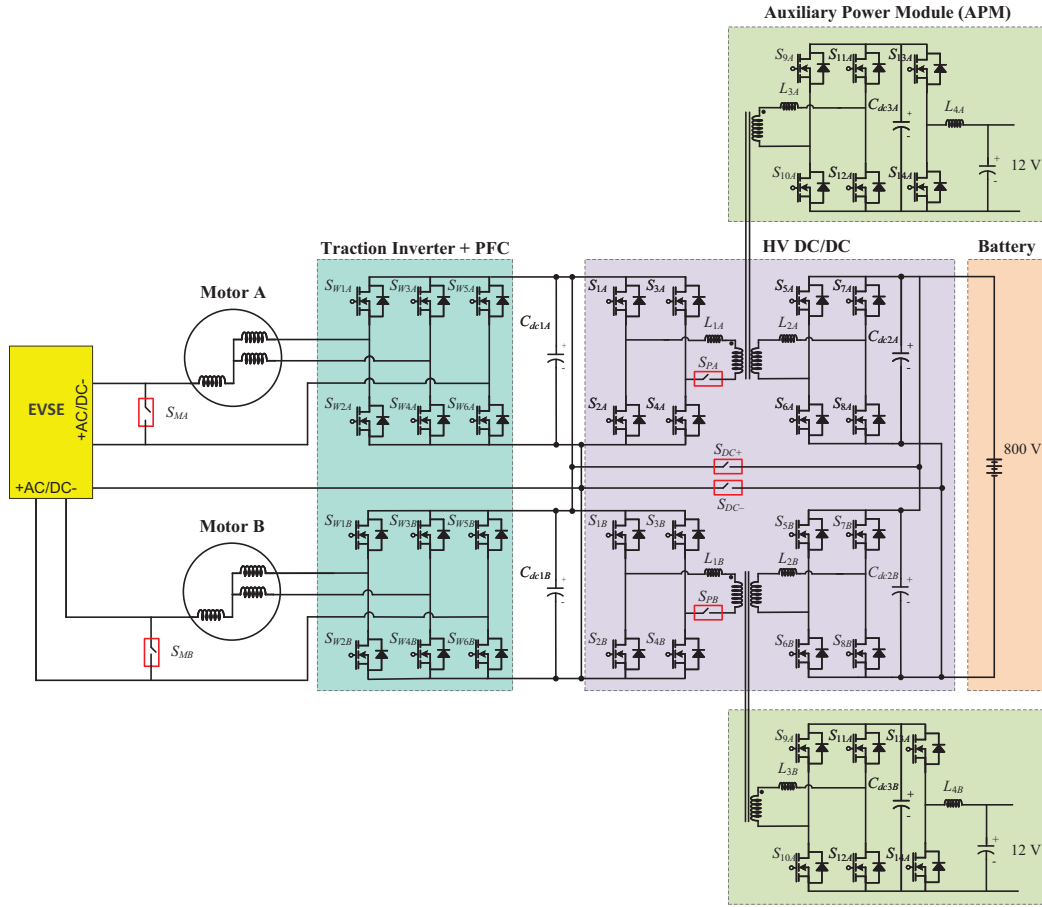


Fig. 3. Diagram of the proposed SuperBox iOBC system for dual motor BEVs.

inverter as a totem-pole circuit, the entire PFC system in Fig. 2 is eliminated. Furthermore, the TAB converter magnetically couples the HV and LV batteries, eliminating the need for the primary H-bridge of the APM shown in Fig. 2, along with its input capacitor. Such system savings are at the cost of introducing S_M and S_P , however, their added cost and size still yield a positive gain in terms of cost, volume, and weight.

C. SuperBox: Modes of Operation

The proposed system operates under three operating modes, as per the truth table of the mode selector relays given in Table I. The three operating modes cover four functionalities: (i) traction, (ii) single/split-phase AC charging, (iii) LV battery charging via APMs, and (iv) DC boost charging.

In traction mode, the $S_{DC+,DC-}$ relays (can be implemented as a single double-pole single-throw switch) connect the HV battery to the common DC link of the parallel inverters. Each TAB galvanically links the DC-link (Port 1), HV battery (Port 2), and LV battery (Port 3). Relays $S_{PA,PB}$ from banks A and B, respectively, disconnect Ports 1 of the TABs, while Ports 2 and 3 operate as a dual active bridge (DAB), enabling bidirectional power flow between the HV and LV batteries.

In AC charging mode, each traction inverter is operated as a two-phase interleaved PFC circuit, utilizing motor winding

TABLE I
RELAY POSITION FOR DIFFERENT MODES OF OPERATION

Mode of Operation	S_M	$S_{DC+,DC-}$	S_P
Traction	1	1	0
AC charging	0	0	1
DC boost charging	0	1	0
Isolated V2X	1	0	1
Non-isolated V2X	1	1	0

inductance as input chokes. The TAB transfers the DC-bus power to charge both HV and LV batteries and enables reverse power flow for vehicle-to-everything (V2X) functionality.

In DC boost charging mode, the 800-V battery can be charged with a legacy 400-V DC charger. Motor windings are configured such that Phase U of each motor is connected in series with phases V and W , forming two interleaved branches of the DC boost converter operating in continuous conduction mode (CCM). The dual-bank configuration provides redundancy and enables higher DC charging power. Port 1 of the TAB is disconnected, while Ports 2 and 3 form a DAB circuit, allowing LV battery charging during DC boost mode.

SuperBox employs Silicon Carbide (SiC) switches, enabling bidirectional power flow. Owing to the dual-bank architecture, SuperBox features three possible output voltage configurations

in V2X mode: (i) 2x120-V single-phase outputs, (ii) 2x240-V split-phase outputs, or (iii) a combination of both.

Additionally, the presence of relays S_P , S_{DC+} , and S_{DC-} allows the V2X system to be configured as either isolated or non-isolated. For isolated V2X operation, relays S_P are closed, while S_{DC+} and S_{DC-} are open. Therefore, power is transferred through the DAB converter. In non-isolated operation, relays S_P are open, and S_{DC+} and S_{DC-} are closed, thereby allowing the power flow to bypass the DAB converter.

III. CHALLENGES OF INTEGRATION

The high level of integration illustrated in Fig. 3 offers significant cost and volume reductions, yet introduces several technical challenges. This section explores some of these challenges and proposes potential solutions.

A. Power Module Dynamic Switching

As discussed in the previous section, the totem-pole circuitry for the PFC circuit is realized using the SiC power module of the inverter. Such a power module is typically optimized for a relatively lower switching frequencies and high currents in traction mode. In other words, a larger die size is implemented in such power modules leading to very low turn-on resistance, $R_{DS(on)}$ but high gate charge, Q_g . Conversely, the PFC operation, especially with motor windings utilized as input chokes, requires high switching frequency in order to establish sufficient choke impedance that yields low current total harmonic distortion (THD). Therefore, optimizing the gate driver is essential for achieving adequate traction and charging efficiency.

The employment of an adjustable gate driver is deemed a key enabler for this technology. For instance, the UCC5880 gate driver by TI offers two gate resistors for both turn-on and turn-off. At a system level, these resistors can be enabled individually or used simultaneously in parallel. Fig. 4 depicts the switching characteristics of a SiC six-pack power module (1200 V/200 A) in LTSpice when driven by the UCC5880 under various driver strengths. With a nominal switching frequency of 10 kHz and 50 kHz for traction and charging modes, respectively, and utilizing a dynamic gate driver, adequate performance of both modes can be achieved.

B. Charging Torque and Elimination Strategy

Another challenge in charging through motor windings is the development of charging torque. Since current is injected into one motor phase and extracted from the other two, the resulting current imbalance leads to electromagnetic torque. This torque presents a safety concern, as it can cause motion when the vehicle is supposedly parked. Even if the induced torque is insufficient to produce motion, it can still result in noise and vibration issues. Therefore, implementing charging torque elimination is essential.

The charging torque expression for an interior permanent-magnet synchronous motor (IPMSM) can be defined as:

$$T_e = \frac{3}{2} P_p i_q (\psi_f + (L_d - L_q) i_d) \quad (1)$$

where P_p is the number of pole pairs, ψ_f is the permanent-magnet flux linkage, respectively, $i_{d,q}$ and $L_{d,q}$ are the d - and q -axes currents and inductances, respectively.

From (1), $T_e = 0$ for $i_q = 0$. Therefore, a control strategy regulating $i_q^* = 0$ is employed [17]. Considering AC current is feeding Phase C, and using Park's transformation and Kirchhoff's current law (KCL) along with (1), the three-phase reference currents, $i_{a,b,c}^*$ yielding $i_q = 0$ are defined as:

$$i_c^* = i_g = \sqrt{2} I_g \cos \theta_g \quad (2a)$$

$$i_b^* = i_c^* \left[\frac{\sin \theta_r - \sin(\theta_r + 2\pi/3)}{\sin(\theta_r - 2\pi/3) - \sin \theta_r} \right] \quad (2b)$$

$$i_a^* = -(i_b^* + i_c^*) \quad (2c)$$

where θ_r and θ_g are the rotor and grid PF angles, and I_g is the RMS grid charging current. Fig. 5 presents the proposed control strategy for charging torque elimination. An outer loop regulates the output HV battery voltage, V_{batt} or current using a PI controller. The output of the PI controller provides i_c^* , which is used to calculate $i_{a,b}^*$ using (2). A simple hysteresis controller is employed to generate the gate pulses for the switches. Note that, i_b^* is undefined for $\theta_r = 5\pi/6 \pm \pi$. Hence, a power derating and denial zones must be established to prevent excessive current [18].

C. Triple Active Bridge Design and Control

The huge voltage difference between the HV and LV batteries usually results in a very high transformer turn ratio

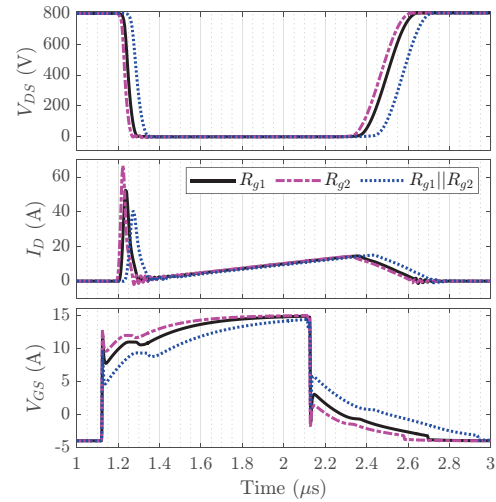


Fig. 4. Switching behavior the FS05MR12A6MA1B SiC power module with a dynamic gate driver ($R_{g1} = 3 \Omega$ and $R_{g2} = 1 \Omega$).

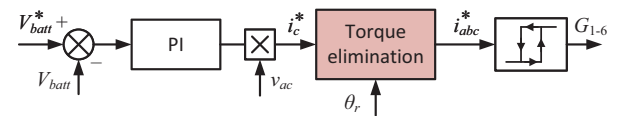


Fig. 5. Diagram of the proposed charging torque elimination control.

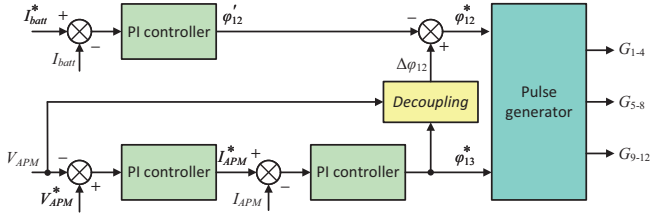


Fig. 6. Proposed power decoupling control scheme for the TAB converter.

if Port 2 were to directly connect to the LV battery. Instead, the voltage of Port 3 is regulated at 48 V, which serves as the input for charging a 48-V battery or supplying power to any 48-V load(s). A synchronous buck converter, as shown in Fig. 3, is used to step down the voltage to 12 V, facilitating the charging of a 12-V battery or powering 12-V load(s). To fully exploit the magnetic integration of the 3PT, the leakage inductance at all three ports should be optimized in a way that can integrate them in the transformer package. This was investigated in an earlier study [8]. Another challenging aspect is the power flow control.

Since the TAB converter is involved in many modes of operation (Table I), an adequate decoupling control must be employed. Fig. 6 shows the proposed control scheme for power decoupling of the TAB converter, adapted from [19]. A closed-loop control regulates the output current at Port 2, I_{batt} using a PI controller, and provides the phase-shift command, φ'_{12} for Port 2. For Port 3, a cascaded control is applied, whereby the outer loop regulates the output voltage, V_{APM} at 48 V and the inner loop controls the output current, I_{APM} that provides the phase-shift command, φ'_{13} for Port 3. A decoupling phase-shift angle, $\Delta\varphi_{12}$, defined in (3), is applied to φ'_{12} to generate the reference phase-shift angle between Ports 1 and 2, φ'_{12} .

$$\Delta\varphi_{12} = \frac{a\pi + 2\varphi_{13} - \sqrt{a^2\pi^2 + \varphi_{13}^2(4 - 4a)}}{2a} \quad (3)$$

where $a = 1 + 1/d$, and $d = (V_3^i L_{12}) / (V_1 L_{23})$ with $L_{12,23}$ are the equivalent leakage inductance linking Ports 1 and 2 and Ports 2 and 3 in the TAB converter, respectively. The $\Delta\varphi_{12}$ enables phase current adjustments between Ports 1 and 3 without interference from Port 2.

IV. SIMULATION RESULTS AND DISCUSSIONS

Subsystem simulations of the proposed SuperBox (Fig. 3) are conducted in MATLAB and PLECS to evaluate the performance in some of the key operating conditions. Due to computational burden, a single bank is evaluated only. The employed motor in this paper is the rear IONIQ-5 IPMSM, whose ratings are listed in Table II along with the SuperBox parameters. The motor is modeled using finite element analysis (FEA) and implemented in PLECS Blockset using its dq flux-linkage and inductance (self and mutual) lookup tables (LUTs). The traction/totem-pole circuit is built using a 1200 V/200 A six-pack SiC FS05MR12A6MA1B power module by Infineon.

TABLE II
SUPERBOX SYSTEM PARAMETERS

Subsystem	Parameter	Symbol	Value	Unit
IONIQ 5 rear motor	Max power	P_{max}	160	kW
	Max current	I_{max}	230	Arms
	Max torque	$T_{e,max}$	350	N·m
	Max speed	$\omega_{r,max}$	15	kRPM
	Pole pairs	P_p	4	—
	PM flux linkage	ψ_f	133	mWb
	Stator resistance	R_s	29	mΩ
PFC (per bank)	Rated power	P_{rated}	9.6	kW
	Output capacitance	C_{dc1}	1500	μF
	Switching frequency	f_s	50	kHz
TAB (per bank)	Port 1 rated voltage	V_1	800	V
	Port 2 rated voltage	V_2	800	V
	Port 3 rated voltage	V_3	48	V
	Turns ratio	$n_1:n_2:n_3$	16:16:1	—
	Switching frequency	f_s	100	kHz
	Dead time	t_d	150	ns
	HV output capacitance	C_{dc2}	20	μF
	LV output capacitance	C_{dc3}	100	μF

A. Charging Torque Elimination

The charging torque elimination control in Fig. 5 is implemented and compared to the conventional totem-pole control at various rotor positions. An AC source is connected across Phase C while the output voltage is regulated to 800 V. Fig. 7 depicts the PFC waveforms at charging power of 9.6 kW and 0° rotor angle. At $t \leq 0.2$ s, interleaved totem-pole control is applied. The resulting RMS charging torque is 18 N·m (Fig. 7(c)). At $0.2 < t \leq 0.5$ s, charging torque elimination is applied, which effectively regulates i_q to zero. Resultantly, the RMS charging torque is only 1.3 N·m. Therefore, the control strategy eliminates the charging torque effectively.

A rotor sweep is conducted to evaluate the charging torque at different rotor positions, as presented in Fig. 8. The RMS torque is as high as 2.4 N·m at $\pi/6 \pm k\pi/2$ with $k \in \{1, 2, 3, \dots\}$ and as low as approximately 0 N·m at $\pi/3 \pm \pi$. In a BEV with park lock or auto park functionality, a desired rotor position can be set prior to charging for minimal torque.

B. PFC Efficiency

The efficiency of the proposed PFC system in SuperBox (i.e., charging through motor windings and six-pack power module) is investigated in this subsection. The semiconductor losses are evaluated in PLECS using the thermal model of device obtained from the datasheet. Motor losses are computed by exporting the current waveforms obtained in PLECS and feeding them into the motor model in Ansys Maxwell. The efficiency of the proposed system is benchmarked against a conventional totem-pole PFC employing 2x493 μH choke inductors and discrete SCT070HU120G3AG SiC devices rated at 1200 V/30 A/ 63 mΩ by STMicroelectronics. Fig. 9 presents the losses incurred by the PFC system during split-phase AC charging at 9.6 kW. Due to rotor saliency, different efficiency of PFC is expected in SuperBox at different rotor positions. As such, best- and worst-case scenarios, at 60° and 120° rotor positions, respectively, are shown in Fig. 9. In the best-case scenario, SuperBox incurs 40% more losses (1.4% less

efficient at 9.6 kW). Assuming a 60 kWh battery, charging at 9.6 kW from 10% to 90% state-of-charge, charging time is extended by approximately 4.6 min. In the worst-case scenario, SuperBox results in a 26 min increase in charging time.

Note that the best-case efficiency scenario occurs at the best-case RMS charging torque. Therefore, 60° is deemed to be optimum rotor angle.

The dynamic gate strength, demonstrated in Fig. 4, was not incorporated in the thermal model of the device used in PLECS. It will be investigated in a later work to close the

efficiency gap between SuperBox and the conventional PFC.

C. Triple Active Bridge Power Decoupling

The TAB operation, charging both HV and LV batteries, is evaluated in this subsection using the control scheme in Fig. 6. A DC source at 800 V is modeled at the inputs of Ports 1 and 2, and a resistive load is used at Port 3 terminals. Fig. 10 demonstrates the TAB operation, where 9 kW of charging power is fed to Port 2 and 1.1 kW, initially, is fed to Port 3. At $30 \mu\text{s}$, a +100% load change is applied at Port 3. The implemented control strategy maintains voltage and current regulation with adequate dynamic response, featuring $20 \mu\text{s}$ settling time. More importantly, Port 2 current regulation is hardly impacted by the load disturbance on Port 3, thanks to the decoupling power flow control.

V. BENCHMARKING

A third-party cost model is employed to conduct an incremental analysis of SuperBox compared to a conventional non-integrated solution. Fig. 11 presents the power stage cost breakdown for both solutions. By eliminating the dedicated totem-pole PFC circuitry and the primary H-bridge for the APM, SuperBox achieves a 7% in the semiconductor costs. The predominant cost factor in semiconductors is the power module, which is present in both systems. Additionally, SuperBox reduces the costs of passive components and transformers thanks to the magnetic integration of the 3PT. However, the cost of relays is higher in SuperBox because S_M and S_P are required for mode selection. The S_{DC+} and S_{DC-} relays are required in both systems to enable 400-to-800 V DC boost charging. Overall, SuperBox achieves 15% cost savings in the power stage. This incremental cost reduction is expected to be significantly greater when considering the mechanical bill of materials owing to the elimination or reduction of connectors, cabling, printed circuit board assemblies (PCBAs), and busbars.

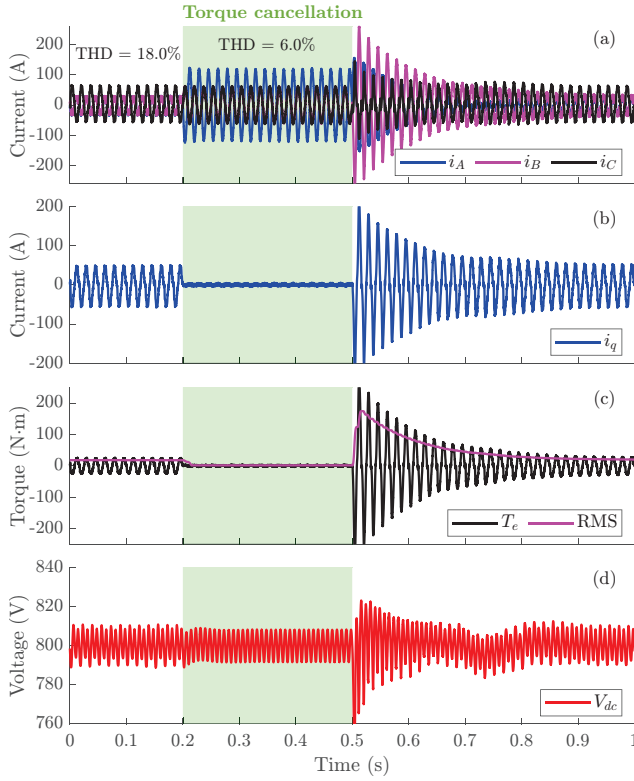


Fig. 7. Simulation results for charging torque cancellation control applied between 0.2 s and 0.5 s, and conventional two-interleaved totem-pole control elsewhere with motor rotor at 0° . (a) $i_{A,B,C}$. (b) i_q . (c) T_e . (d). V_{dc} .

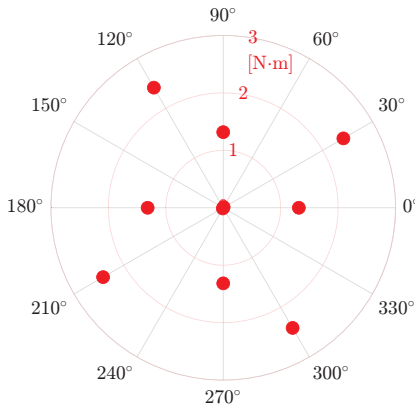


Fig. 8. RMS torque vs. rotor angle under charging torque cancellation control at 9.6 kW split-phase AC charging.

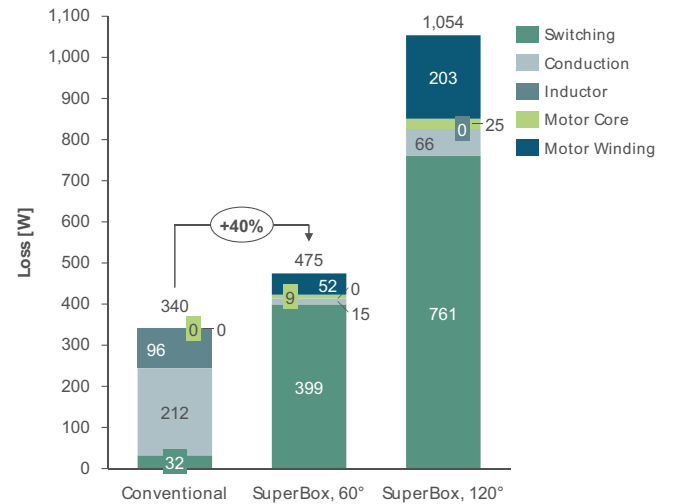


Fig. 9. Losses at 9.6 kW split-phase AC charging in conventional PFC stage and SuperBox at 60°C switch (junction) and motor temperatures.

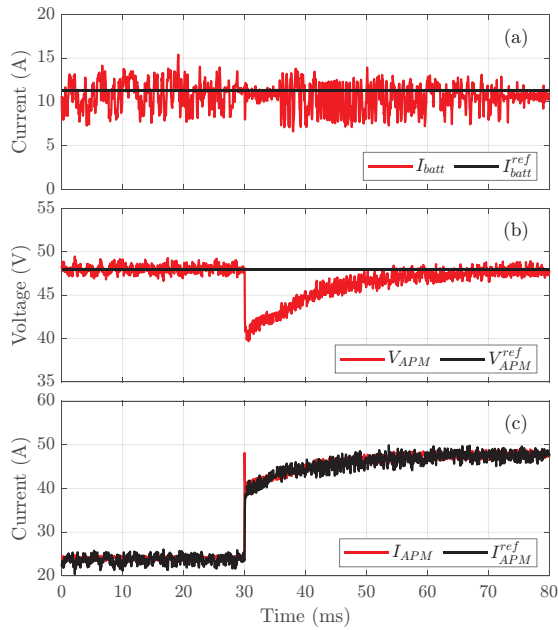


Fig. 10. TAB operation with power decoupling control at 9 kW and 1.1 kW at Ports 2 and 3, respectively, with +100% load change at Port 3 at 30 μ s. (a) Port 2 current. (b) Port 3 voltage. (c) Port 3 current.

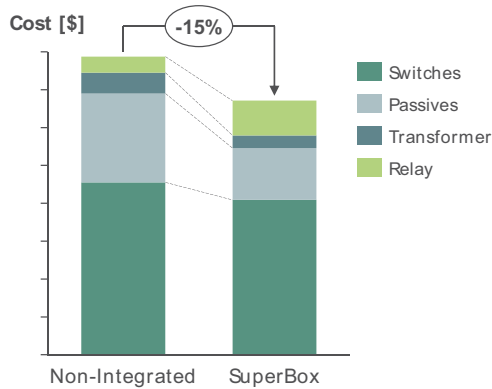


Fig. 11. Power stage cost breakdown of SuperBox compared to conventional non-integrated solution.

VI. CONCLUSION

The proposed SuperBox system introduced a highly integrated approach to power electronics conversion for EVs. Different power electronics conversions within the EV are consolidated into a single unit, saving system costs and reducing volume. First, a standardized motor is considered as the electric machine, employing the three-phase Y-connected motor windings to serve as the PFC coil for the front-end PFC converter of the OBC and as inductors for an interleaved boost converter for DC boost charging. Second, an integrated three-port transformer for DC/DC conversion is proposed with a three-port architecture. Third, dual bank architecture provides system redundancy. Challenges related to this level of system integration, such as PFC/traction control, charging torque elimination, and DC/DC power decoupling, were investigated,

and their potential solutions were demonstrated through simulation. Power stage benchmarking suggests 15% cost reduction compared to conventional non-integrated system. Future work will focus on validating DC boost and traction modes.

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